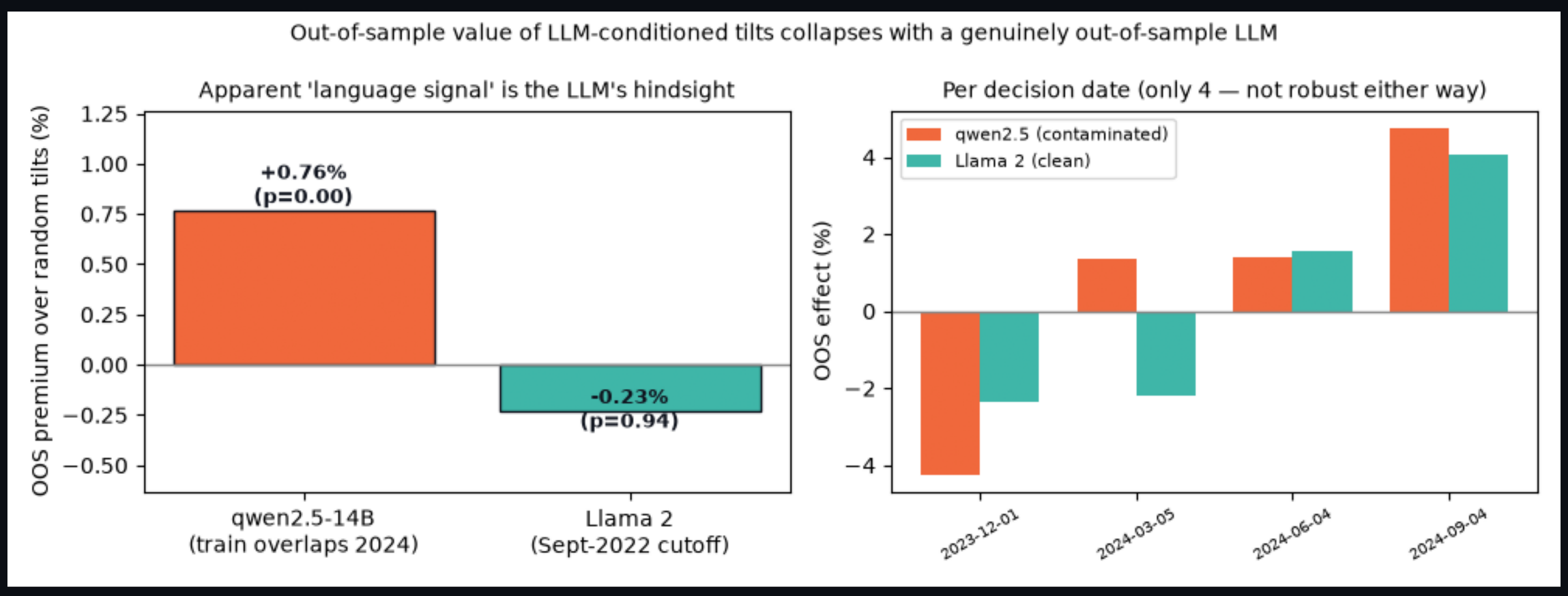
Does Language Really Help a Quantum Portfolio Optimizer?

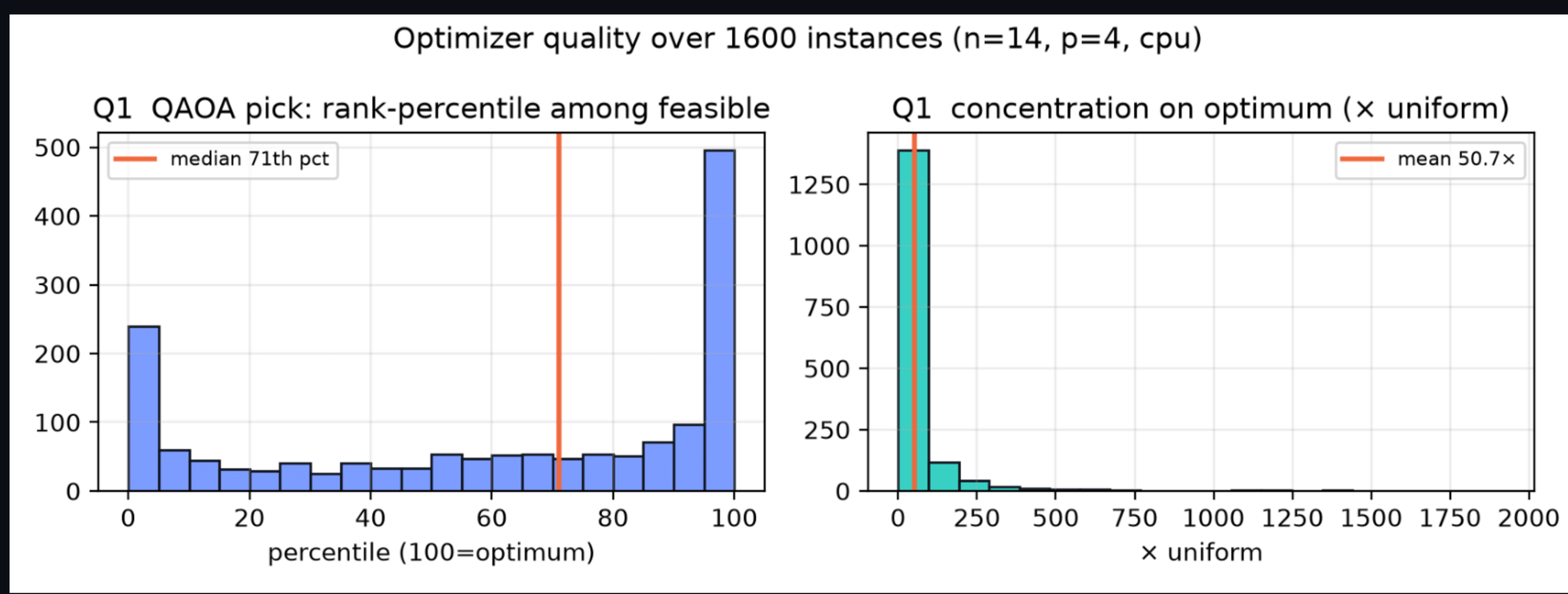
A Bias-Controlled Study and a Training-Horizon Confound in LLM-Conditioned QAOA

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QUESTION When a backtest says 'LLM news-views help a quantum portfolio optimizer,' is that signal — or an artifact? We build a deliberately bias-controlled pipeline to catch ourselves.

PIPELINE Objective universe (Dow 30, CRSP total returns) -> an LLM reads RavenPack headlines time-stamped BEFORE each decision date into per-asset convictions -> a +-10% view tilts the QAOA cost Hamiltonian -> hold 5 of 14, evaluate the held-out next quarter. CONTROLS: matched random-tilt placebo; ablation vs RavenPack's own sentiment; and TWO LLMs on the SAME headlines — qwen2.5-14B (training overlaps 2024) vs Llama 2 (Sept-2022 cutoff, genuinely out-of-sample).





KEY FINDING — the apparent 'language signal' is the LLM's hindsight

qwen2.5 (train overlaps 2024): +0.76% OOS premium over random tilts (p=0.00)

Llama 2 (clean, pre-2023): -0.23% (p=0.94) — indistinguishable from random

=> the out-of-sample 'alpha' VANISHES with a genuinely out-of-sample LLM.

SUPPORTING language has real leverage (changes 68% of optima), but shallow QAOA is a WEAK exact optimizer — finds the true optimum only 7.7% of the time. No quantum-advantage claim.

HONEST LIMITS only 4 decision dates (instance-level p pseudo-replicated); small-n exact sim, no hardware; large-cap US; no transaction costs; additive tilt. None inflate the central result.

CONTRIBUTION a reproducible bias-control template + a quantified warning: date your LLM against your backtest, or training-horizon look-ahead will manufacture alpha that isn't there.

Reproducible: python -m paper.study.run_study / control / compare_figure · fixed seeds, committed CRSP + conviction caches · scales on HPC (SLURM/GPU)